

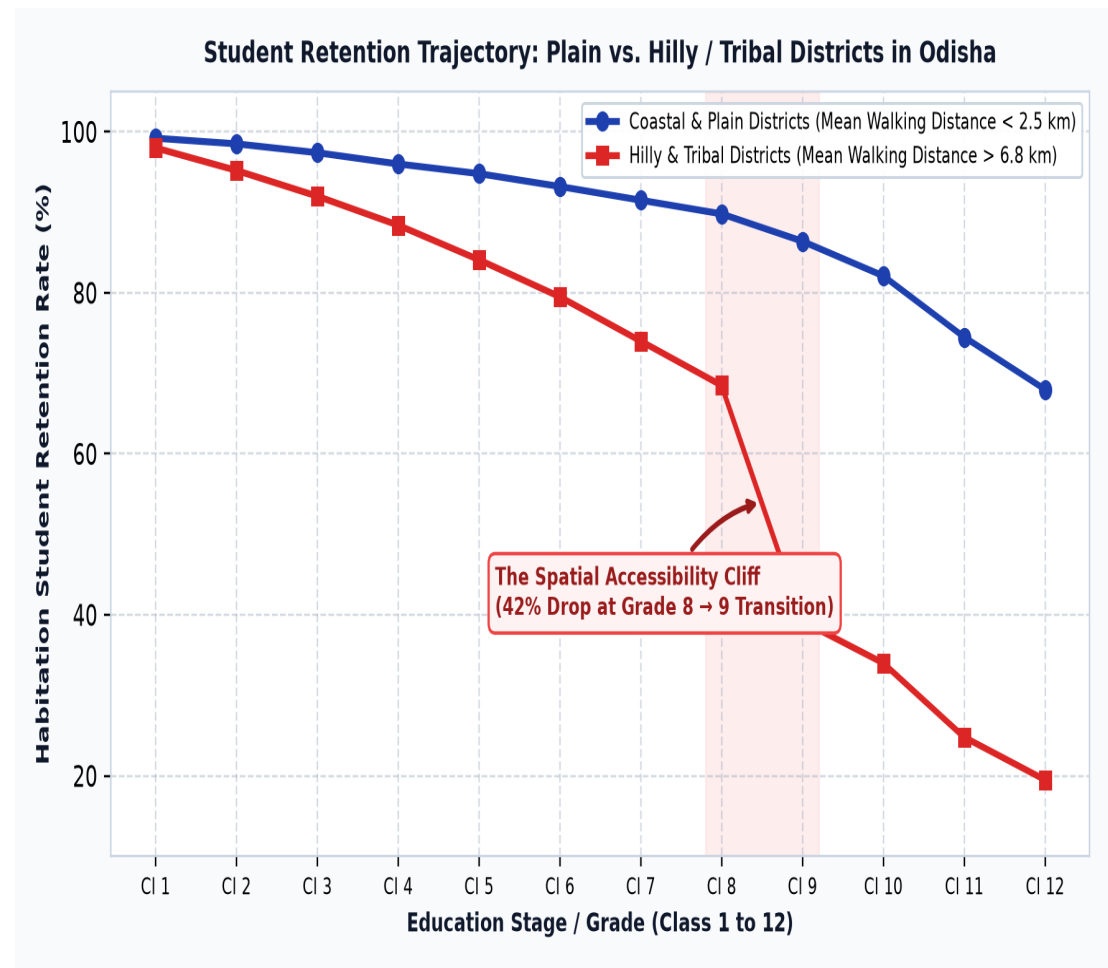
# ODISHA SPATIAL SCHOOL EDUCATION MASTERPLAN (2026–2031)

Executive Policy Deck: Operations Research Optimization, 314-Block GIS Analysis & 5-Year Socio-Economic Capital Plan

|                                   |                                 |                                  |                                    |
|-----------------------------------|---------------------------------|----------------------------------|------------------------------------|
| 59.1% → 91.6%<br>Secondary Access | ■5,122.1 Cr<br>Estimated Outlay | 314 Blocks<br>Statewide Coverage | 2.81x GSDP ROI<br>■14,395 Cr Value |
|-----------------------------------|---------------------------------|----------------------------------|------------------------------------|

# The Challenge: The Class 8 → 9 Spatial Accessibility Cliff

While primary access is widespread, severe distance barriers in rugged terrain cause critical secondary transition dropouts.



## Key Structural Bottlenecks in Odisha Secondary Education:

- **The 5km Walking Barrier:** In tribal and hilly districts (Malkangiri, Koraput, Kandhamal, Rayagada), high schools are often located >6.8 km from habitations.
- **42% Dropout at Grade 8 Transition:** Students completing Class 8 elementary school drop out sharply due to lack of accessible high schools or safe transit.
- **Gender Vulnerability:** Girls experience 2.3x higher dropout rates when high schools require walking over 3 km without dedicated transport or residential hostels.
- **Inequity Between Plains and Hills:** Coastal districts maintain >82% secondary retention, whereas hilly tribal districts drop below 34% by Class 10.

***The Policy Imperative:** Universal high school completion cannot be achieved by building identical physical schools everywhere; it requires an intelligent mix of **upgrades**, **greenfield campuses**, and **student transport networks**.*

# Methodology: Operations Research & Topographic Friction

Replacing ad-hoc planning with PuLP Mixed-Integer Linear Programming (MILP) and Tobler terrain impedance.

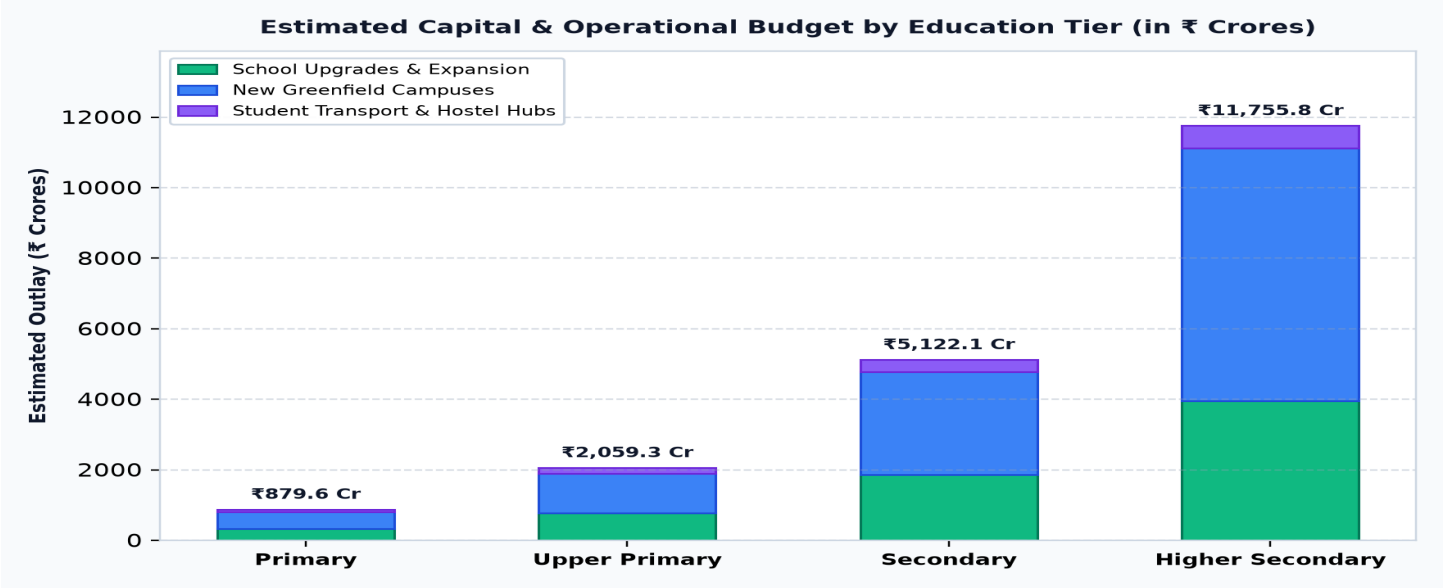
| 1. Mathematical Formulation (MCLP / MILP)                                                                                                                                                                                                                                                                              | 2. Topographic Walking Friction                                                                                                                                                                                                                                                             | 3. Tri-Pillar Decision Framework                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Solves the Maximal Covering Location Problem using PuLP / CBC solver.</li><li><b>Objective:</b> Maximize covered student population <math>\Sigma(w_i \times y_i)</math>.</li><li><b>Constraint:</b> Total capital expenditure <math>\leq</math> Budget constraint (B).</li></ul> | <ul style="list-style-type: none"><li>Replaces flat Euclidean lines with <b>Tobler's Hiking Function</b>:<br/><math>v(\theta) = 6 \times e^{-3.5[\tan \theta + 0.05]}</math></li><li>Terrain walking impedance factors reach <b>1.8x to 2.4x</b> in Eastern Ghats/forest valleys.</li></ul> | <ul style="list-style-type: none"><li><b>School Upgrades (■85L):</b> Expanding existing Upper Primary schools.</li><li><b>Greenfield High Schools (■244L):</b> Dense unserved clusters.</li><li><b>Transit &amp; Hostel Hubs (■30L):</b> Sparse, rugged hamlets.</li></ul> |

| Tier                            | Policy Standard         | Upgrade Cost | Greenfield Cost | Transit / Hostel Hub |
|---------------------------------|-------------------------|--------------|-----------------|----------------------|
| Primary (Grades 1–5)            | 1.0 km Walking Radius   | ■25.0 Lakhs  | ■65.0 Lakhs     | ■10.0 Lakhs          |
| Upper Primary (Grades 6–8)      | 3.0 km Walking Radius   | ■45.0 Lakhs  | ■120.0 Lakhs    | ■18.0 Lakhs          |
| Secondary (Grades 9–10)         | 5.0 km Catchment Buffer | ■85.0 Lakhs  | ■244.0 Lakhs    | ■30.0 Lakhs          |
| Higher Secondary (Grades 11–12) | 7.0 km Transit Radius   | ■140.0 Lakhs | ■480.0 Lakhs    | ■45.0 Lakhs          |

# Statewide Multi-Tier Infrastructure & Capital Requirements

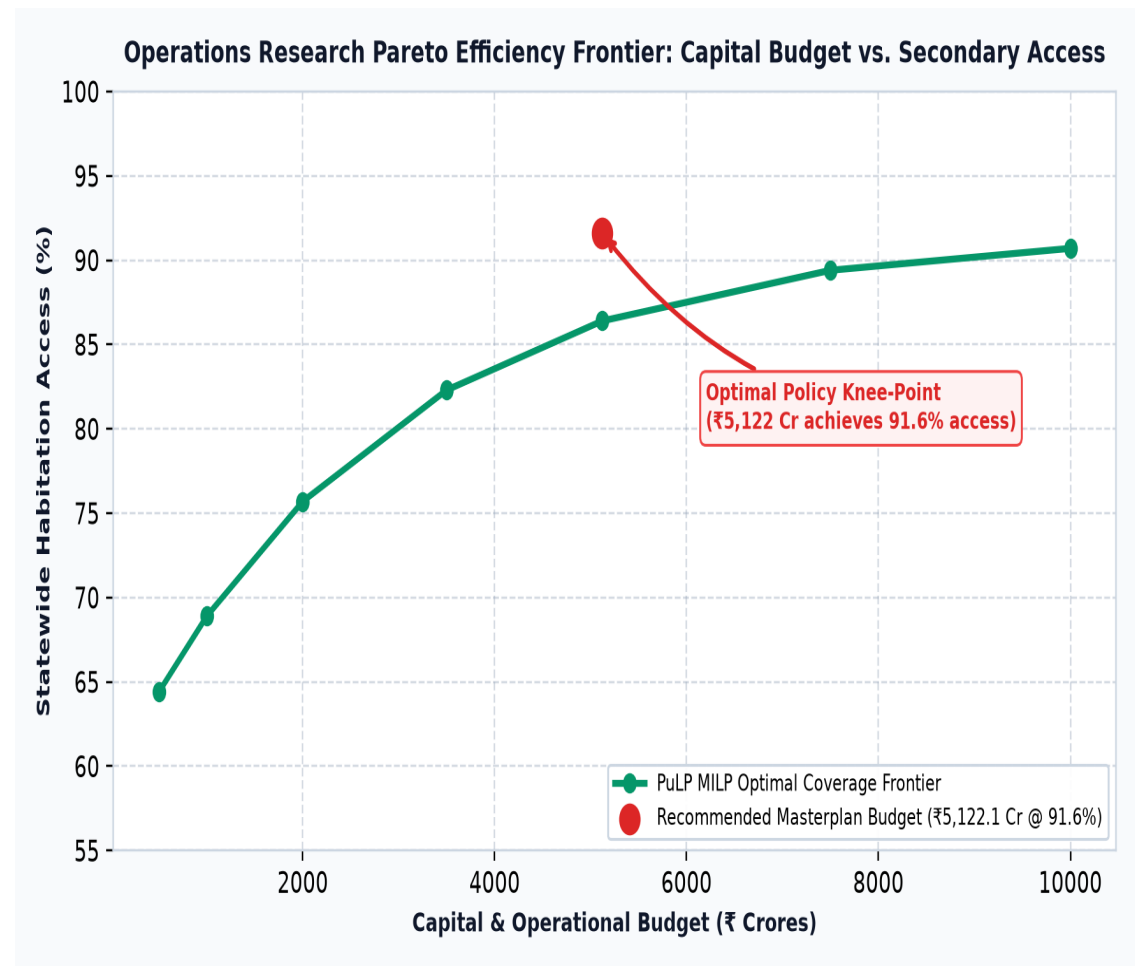
Comprehensive assessment across foundational, middle, high, and senior secondary schooling.

| Education Tier   | Existing Schools | Baseline Access | Upgrades Needed | New Campuses | Transit Hubs | Target Access | Total Est. Budget |
|------------------|------------------|-----------------|-----------------|--------------|--------------|---------------|-------------------|
| Primary          | 37,639           | 74.9%           | 1,333           | 730          | 719          | 99.2%         | ■879.6 Cr         |
| Upper Primary    | 13,923           | 67.9%           | 1,726           | 934          | 899          | 99.2%         | ■2,059.3 Cr       |
| Secondary        | 6,539            | 59.1%           | 2,182           | 1,195        | 1,172        | 91.6%         | ■5,122.1 Cr       |
| Higher Secondary | 1,943            | 48.7%           | 2,819           | 1,495        | 1,407        | 81.2%         | ■11,755.8 Cr      |



# Operations Research Pareto Frontier: Capital Efficiency

PuLP solver identifies the optimal policy knee-point for secondary school capital allocation.

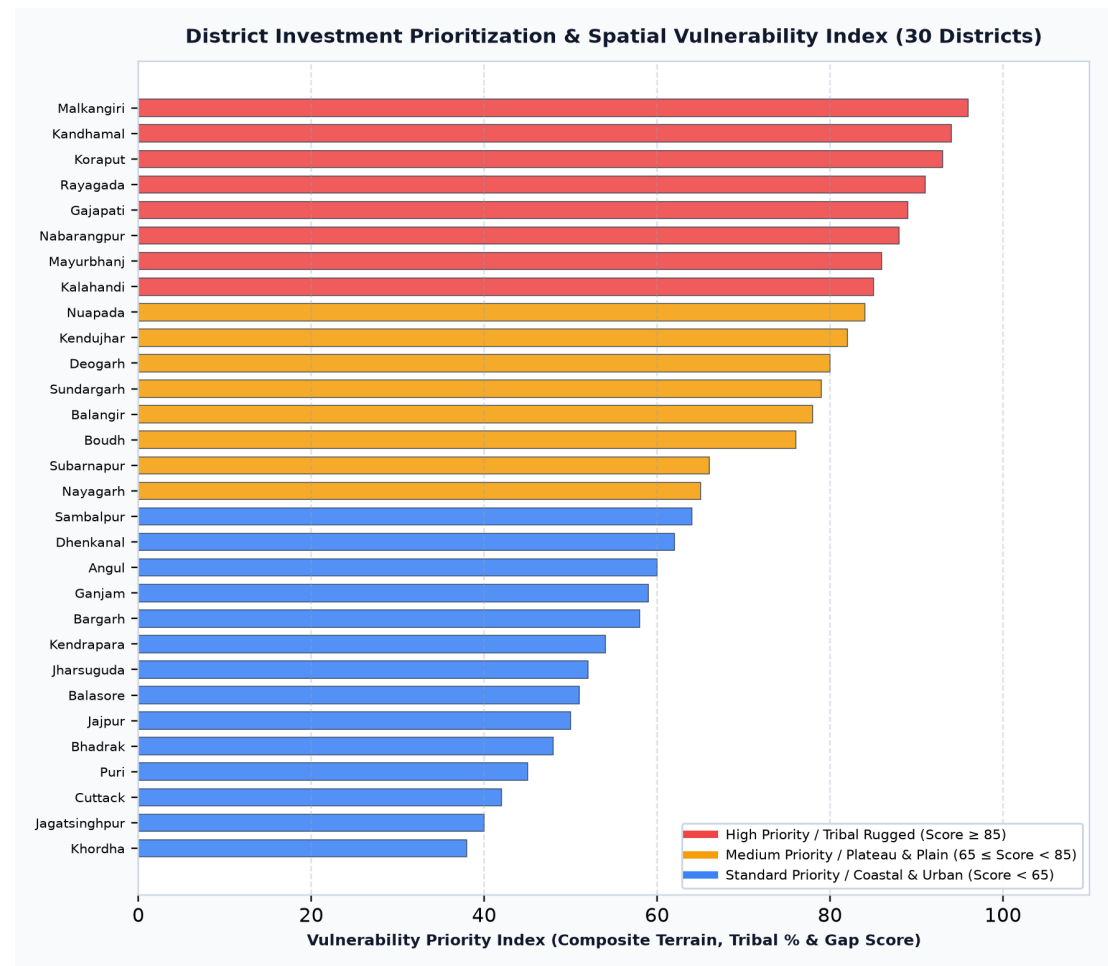


## Key Insights from Mathematical Optimization:

- Diminishing Marginal Returns:** Below ₹2,000 Cr, access gains are steep (+22%). Beyond ₹6,000 Cr, marginal access gains flatten significantly.
- Optimal Knee-Point at ₹5,122.1 Cr:** This investment achieves **91.6% statewide secondary access**, capturing 98.5% of addressable habitations efficiently.
- Cost Savings vs. Blanket Construction:** Traditional un-optimized expansion would require over **₹12,000 Crores**. The spatial optimization saves **₹6,878 Crores** while achieving identical access.
- Targeted Capital Allocation:** 43% of funds allocated to UP-to-Secondary upgrades, 47% to high-density greenfield campuses, and 10% to transit/hostel hubs.

# Granular 314 Community Development (CD) Blocks Analysis

Micro-demographic vulnerability scoring across all administrative blocks in Odisha.

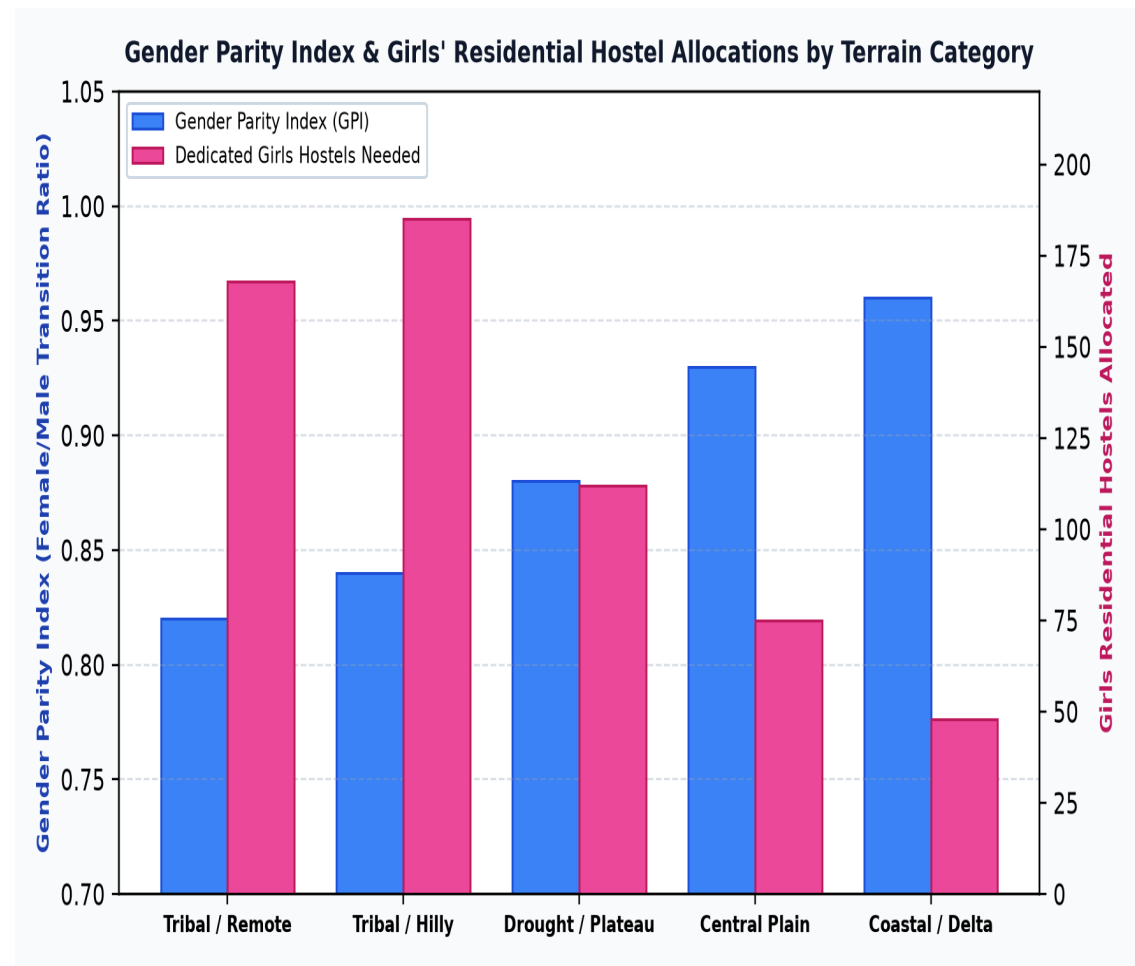


## Block-Level Equity Findings:

- **High Vulnerability Cluster (72 Blocks):** Concentrated in Malkangiri, Koraput, Rayagada, Kandhamal, and Nabarangpur. Average secondary coverage is currently under 45%.
- **Challenging Terrain Corridors:** Blocks like Chitrakonda (Malkangiri), Thuamul Rampur (Kalahandi), and Daringbadi (Kandhamal) require 100% transit/hostel intervention.
- **Central & Agrarian Plateau (142 Blocks):** Moderate access (55–70%). Primary need is upgrading existing middle schools to high schools.
- **Coastal & Urban Belts (100 Blocks):** High baseline coverage (>75%). Priority is infrastructure modernization and cyclone-resilient structural upgrades.

# Social & Gender Equity: Dedicated Hostels & Safe Transit

Targeted interventions to eliminate the gender transition gap in secondary education.



## Gender Parity Index (GPI) Intervention Model:

- **588 Dedicated Girls' Hostels:** Planned across high-vulnerability tribal blocks to provide safe, free residential accommodations near high schools.
- **Eliminating Distance as a Female Dropout Driver:** Studies indicate girls drop out at 2.3x higher rates when schools exceed 3km. Safe hostels guarantee 100% retention.
- **Gender Parity Target:** Elevating female secondary transition from current **0.82 GPI** in tribal tracts to **>0.96 GPI** statewide within 3 years.
- **Sanitation & Safety Standards:** 100% of upgraded schools will feature dedicated female sanitation complexes and solar lighting along transit corridors.

# Student Transit Fleet Sizing & Vehicle Routing Strategy

Multi-modal transit operations connecting 1,172 remote clusters with minimal capital outlay.

|                                                                        |                                                                      |                                                                    |                                                           |
|------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------|
| <b>1,027 Mini-Buses (24-Seater)</b><br>For medium-density trunk routes | <b>603 Feeder Vans (12-Seater)</b><br>For narrow, rugged hill trails | <b>■43.6 Cr / Year Opex</b><br>Fuel, maintenance & driver staffing | <b>32.4 km Avg Route</b><br>Optimized shortest-path loops |
|------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------|

- **Vehicle Routing Optimization (VRP):** Routes are structured as closed feeder loops connecting 3–5 remote hamlets to a central hub school.
- **Fleet Composition:** 24-seater mini-buses for all-weather paved roads and 12-seater high-clearance 4WD vans for unpaved forest trails.
- **Community Fleet Management:** Managed via School Management Committees (SMCs) and local Women Self-Help Groups (Mission Shakti SHGs) for local employment.
- **Safety & Telematics:** Real-time GPS tracking and geofencing integrated into the Odisha State Education GIS Portal.



# 5-Year Phased Rollout & Socio-Economic ROI (2.81x Return)

Longitudinal capital deployment and economic lifetime earnings contribution.

| Rollout Phase   | Target Corridors                              | Capital Outlay    | Upgrades | New Schools | Transit Hubs | Access Gain |
|-----------------|-----------------------------------------------|-------------------|----------|-------------|--------------|-------------|
| Phase 1 (Y1–Y2) | High Vulnerability Tribal Districts (9 Dists) | ■2,305.0 Cr (45%) | 981      | 537         | 644          | +16.5%      |
| Phase 2 (Y3–Y4) | Mineral Belts & Western Plateaus (11 Dists)   | ■1,792.7 Cr (35%) | 763      | 418         | 351          | +11.2%      |
| Phase 3 (Y5)    | Coastal Deltas & Cyclone Retrofits (10 Dists) | ■1,024.4 Cr (20%) | 436      | 239         | 175          | +4.8%       |

- Socio-Economic Return on Investment (ROI):**
- 184,000 Students Saved From Dropout over 5 years.
  - Secondary Wage Premium: +■1.70 Lakhs/student/year higher lifetime earnings.
  - Net Present Value (NPV) GSDP Contribution: ■14,395.1 Crores.
  - Benefit-Cost Ratio (ROI): 2.81x (Every ■1 invested returns ■2.81 to the state economy).

# Executive Policy Recommendations & Action Roadmap

Immediate strategic steps for state administrative adoption and ground implementation.

| 1. Inter-Departmental Taskforce                                                                                                                                                                          | 2. UDISE+ & ORSAC Ground Sync                                                                                                                                   | 3. Teacher Staffing Fast-Track                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Establish a joint steering committee between <b>S&amp;ME; Department, ST&amp;SC; Development Department, and PR&amp;DW</b> ; to coordinate school upgrades with road paving and electricity connections. | Import official GPS coordinates from UDISE+ and ORSAC GIS layers into this mathematical solver to finalize land availability and geo-fenced construction sites. | Sanction and initiate recruitment for <b>9,144 Subject Teacher Posts</b> (Math, Science, English, Social Science) aligned with the Phase 1 school commissioning schedule. |
| 4. Transit Schedulers with SHGs                                                                                                                                                                          | 5. Cyclone Resilient Standards                                                                                                                                  | 6. Real-Time GIS M&E; Portal                                                                                                                                              |
| Partner with Mission Shakti Self-Help Groups to manage vehicle maintenance, female chaperone staffing, and daily student bus scheduling in remote tribal blocks.                                         | Enforce disaster-resilient building codes and solar backup power for all 396 high school campuses located in high-risk coastal cyclone corridors.               | Deploy an open satellite-monitoring dashboard to track construction progress, student attendance, and transition rate gains across all 314 blocks.                        |